

CN451 Network Programming

Final-Project: Network Simulation and Automation with Python

Dr. Mohamed MUSBAH

By doing the tasks of this project, you should learn how to use Python to

- Use a network simulator to emulate ip network scenarios.
- Use `telnet` and `ssh` approaches to configure and communicate with network devices..
- Use Python to perform simple networking tasks and to acquire device information.
- Use networking python libraries (e.g. `netmiko`) to conduct automation tasks.

Important Note: Do not forget to use (**help**) and (**dir**) commands for more help.

Scenario: In this project, you are asked to develop a network application on a virtual network environment (A simulator of your choice). Also, you are expected to test and save your tests' results on an output text file. The application requirements are:

Project Tasks

Task 1: Use the simulator to build a network scenario (of your choice) that contains; 1 router, 2 switches (from different vendors) and at least 3 hosts (one of them to work as a server). Do the basic configurations to make these devices communicate to each other using consoles and telnets.

Task 2: manually, create three input text files; one for network devices list (contains name and ip address), one for the login information for each device, and the other for some configuration commands (at least two of your choice) for each device. Now, make your main Python script that reads (and validates!!!) information from the text files, then it accesses each device and applies the corresponding commands on them. One of the important tasks of this script is to check the reachability of each ip address in this network.

Task 3: in addition to your scripts, let the application extract the routers' CPU utilization value and save it on an output text file in a real-time manner. Build a graph to show the progress of these CPU values.

Task 4: add a feature to your application using the Scapy packet manipulation programme. This feature does some basic network packet sniffing on a network interface (of user's choice) and shows the requested results.

Task 5: You may want to add extra features of your choice (at least two) to your application using different libraries and tools that those you used in your main programme.

Task 6: You are now asked to perform a network application that runs over this network. Use the application you made in mini-project II (Chat Application) over three client hosts and one server and check how it works.

Important Note: You are expected to design your programme in a modular-fashion, so that you make separate .py files that contains some functions and classes you may want to use in your main programme by importing them.

*** You have complete flexibility in designing the commands, responses, warnings, and error messages for each of the aforementioned tasks, tailoring them as you see fit. ***

Submission Requirements

This could be a team-based project (max two students). If so, please make one submission and show the team members names clearly

You **MUST** submit the following three items via Google Classroom:

- 1- Python Scripts: Your source code as a *.py files.
- 2- Application Report (User Guide-Like): A Google Docs file that documents the whole app and how every component works (No code, just use case and how to use).
- 3- Verification Video (Google Vids LINK): A short (6–8 minutes) screen recording with your voice (and face) [If two students share the project, both must participate!!!!]. The video MUST have a sensible name. In the video, you must:
 - Briefly show your code.
 - Run the script in real-time.
 - Explain your new feature. What it does and how it works?

Submissions without a video will NOT be graded to ensure academic integrity.

Late submissions will NOT be allowed as this is the final project.